testium

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CHAPTER

ONE

OVERVIEW

testium is an automated test framework written in Python by François Dausseur. It uses the Qt6 graphical framework.

It has been developed since 2013 with production and development testing in mind.

Its function is to automate the execution of tests. It can be invoked either as command line application or as a graphical interface application.

Sources and pre-built releases are hosted at github.com/testium-HIL/testium[1].

The quickest installation is from PyPI:

```
pip install testium-hil
```

The PyPI project is named `testium-hil`; the installed command is `testium`. The install includes the GUI and the language server. Self-contained binaries (Linux binary, AppImage, Flatpak, Windows installer) are published on the GitHub releases page.

The tool also generates test reports and lets you customize them.

Its main features are:

- YAML test description,
- Test configuration files in YAML,
- Full range of pre-existing Test items,
- Test steps, loops,
- Dynamic variables expansion at test runtime,
- Conditional test step execution,
- Modularity of tests (reusable test sequences),
- etc.

These features let the test engineer write efficient and robust tests.

Each test is described with the help of a YAML[2] file having .tum as extension. This file is analyzed and then displayed as a tree in the GUI (see Figure above).

[1] https://github.com/testium-HIL/testium
[2] https://yaml.org/

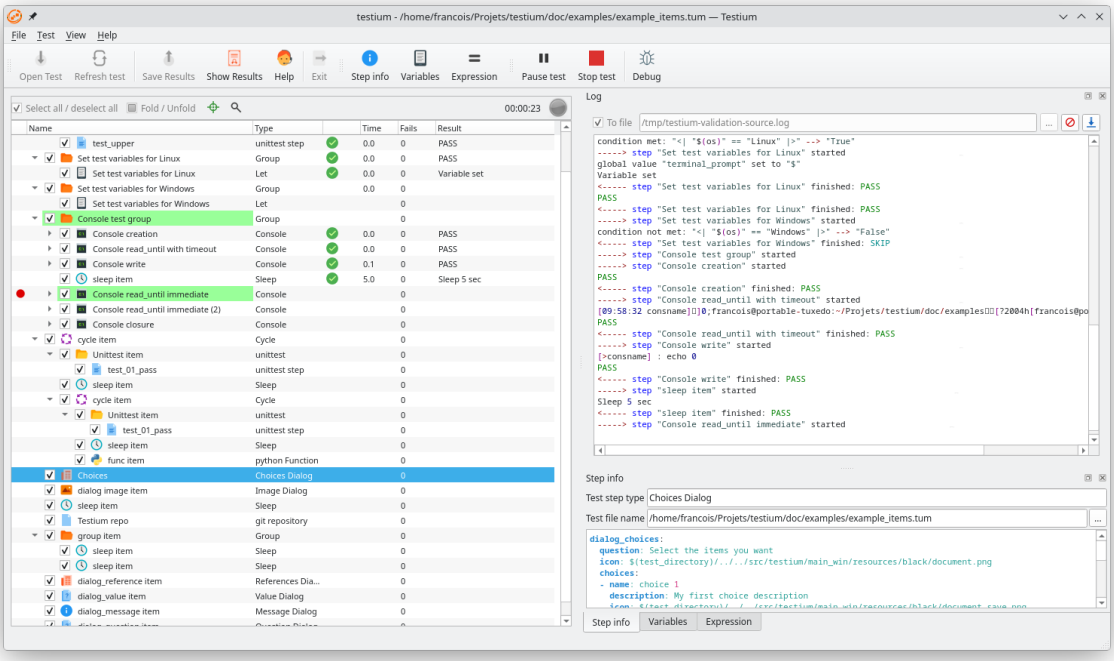


Fig. 1.1: testium

CHAPTER

TWO

COMMAND LINE INTERFACE

```
usage: testium.pyw [-h] [--version] [-b] [-c CONFIG_FILE [CONFIG_FILE ...
↪]] [-r] [-l LOG_FILE]
                   [-d DEFINE [DEFINE ...]] [-p REPORT_FILE] [-t
↪{sqlite,json,junit,html,text}]
                   [-n REPORT_PATTERN [REPORT_PATTERN ...]] [-i␣
↪INCLUDE_PATH [INCLUDE_PATH ...]] [-o] [-g]
                   [test_file]

positional arguments:
test_file             the test script file

optional arguments:
-h, --help            show this help message and exit
--version             Returns the version of testium
-b, --batch-execution
                        Executes the test in batch mode
-c CONFIG_FILE [CONFIG_FILE ...], --config-file CONFIG_FILE [CONFIG_FILE ..
↪.]
-o, --no-color        Deactivates stdout colors in batch mode
                        Configuration file
-r, --run-and-close   Runs the test then closes the application
-l LOG_FILE, --log-file LOG_FILE
                        log file name
-d DEFINE [DEFINE ...], --define DEFINE [DEFINE ...]
                        Configuration passed to the executed tests.
-p REPORT_FILE, --report-file REPORT_FILE
                        report file name
-t {sqlite,json,junit,html,text}, --report-type
↪{sqlite,json,junit,html,text}
                        report file type
-n REPORT_PATTERN [REPORT_PATTERN ...], --report-pattern REPORT_PATTERN␣
↪[REPORT_PATTERN ...]
                        report file pattern
-i INCLUDE_PATH [INCLUDE_PATH ...], --include-path INCLUDE_PATH␣
↪[INCLUDE_PATH ...]
                        Python modules search path
-g, --debug           GUI debug mode
```

2.1 -h, --help

Prints the usage message shown above.

2.2 -b, --batch-execution

Executes the test in text mode. Qt does not need to be installed in this mode.

2.3 -o, --no-color

Disables colored output in batch mode.

2.4 -c, --config-file

This option provides configuration file(s) from the command line. The configuration files format and content is detailed in the *config files* section.

If this parameter is not given while calling *testium*, the default configuration files will be used.

2.5 -r, --run-and-close

This parameter makes testium close immediately after running the `test_file` passed on the command line.

If there is no `test_file` argument passed, this option is ignored.

2.6 -l, --log-file

Path of the log file for the test execution. If not provided, the log is written to a temporary folder.

2.7 -d, --define

Defines one or more variables in the form `VARIABLE1=value1 VARIABLE2=value2 ...`". Then, these variables are available from the test scripts, using the *global variables* *testium* feature. A define overrides a configuration file value; values derived from the overridden variable follow.

2.8 -p, --report-file

Path of the report file, stored during the test execution.

This option is only useful in *batch mode*.

2.9 -t, --report-type

This option is used together with option *-p* and defines the type of report to generate.

Please read the *reports* section for more details on the possible types of report.

2.10 -n, --report-pattern

This option is used in conjunction with option *-p* and defines the report pattern(s) used to filter the report results which will be included in the report file.

More details in *reports* section.

2.11 -i, --include-path

Additional Python paths. These paths are appended to the sys.path[3] python variable.

2.12 -g, --dev-debug

Development option: attaches a debugger to *testium* itself. See `CONTRIBUTING.md` of the source repository. Not used for debugging tests; see the *Debugging your tests* chapter.

[3] https://docs.python.org/3/library/sys.html?highlight=sys%20path#sys.path/

CHAPTER

THREE

MODES OF OPERATION

3.1 Graphical mode

testium was originally designed as a graphical (GUI) application.

Run the `testium` command to start it. This is the default mode.

3.2 Batch mode

Batch mode executes a test in text mode. In this mode, the test does not start any graphical interface.

Listing 3.1: call a test in batch mode

```
testium -b test/my_test/main.tum
```

3.3 Language server (editor support)

testium ships a Language Server Protocol[4] server so that `.tum` files get editor assistance in any LSP-capable editor: completion of test item types, hover documentation of their parameters, and an outline view.

The server speaks LSP over standard input/output and is started with:

Listing 3.2: start the language server

```
testium lsp
```

It is not meant to be launched directly by the user: an editor's LSP client spawns it and drives the exchange. A VSCode / VSCodium client extension, *testium_assist*, is provided for that purpose; any other LSP client (Neovim, Emacs `lsp-mode`, …) can be pointed at `testium lsp` as well.

The information the server exposes is the test item schema, which can also be dumped as JSON for inspection or tooling:

[4] https://microsoft.github.io/language-server-protocol/

Listing 3.3: dump the item / parameter schema

```
testium schema
```

Because the schema is built from *testium* itself, every new item type or parameter becomes available in the editor on the next *testium* upgrade, with no change to the client.

A committed copy of this schema, `schema/tum.json`, is kept at the root of the source repository for tools that read a file instead of running the command (yaml-language-server, external linters, AI assistants).

The language server is installed with *testium* in every channel.

3.3.1 Installing the VSCode / VSCodium extension

The *testium_assist* client extension is published on Open VSX[5], the registry used by VSCodium, Cursor, Windsurf, Eclipse Theia and code-server. In those editors, open the Extensions view and search `testium-assist`, or install it from the command line:

Listing 3.4: install in VSCodium and other Open VSX editors

```
codium --install-extension testium.testium-assist
```

Microsoft *VSCode* uses a different marketplace that does not list Open VSX extensions, so install the packaged `.vsix` by hand. Download it from the Open VSX page linked above, then either choose *Extensions* → ⋯ → *Install from VSIX...* in the UI, or run:

Listing 3.5: install the .vsix in Microsoft VSCode

```
code --install-extension testium-assist-0.1.0.vsix
```

The extension launches `testium lsp`, so the `testium` command must be on the PATH. If *testium* is installed elsewhere (a specific binary or an AppImage), point the `testium.serverPath` setting at it instead.

Once installed, open a `.tum` file: completion of item types, hover documentation and the outline view become available. If nothing happens, check that no `files.associations` entry forces `*.tum` to another language (it must stay the `tum` language the extension provides).

[5] https://open-vsx.org/extension/testium/testium-assist

CHAPTER

FOUR

TUM FILE SYNTAX

testium is a python-based tool which uses a YAML based description file to operate tests: the TUM file.

The description of tests is based on the definition of test sequences. There is a main YAML element which is the *testium* tool entry point.

All other steps are listed in the step list of the main. Some steps, such as loop or console items, are themselves lists of steps.

In YAML, nesting is expressed by indentation.

YAML language auto detects data type so that it is not necessary to cast the element type explicitly. See YAML home page[6] for further information on the language.

The example below shows a basic implementation of the TUM description file:

Listing 4.1: main test file

```
main:
    name: Test example
    steps:
        - test_item1:
            name: test_1
        - loop :
            name: test cycle
            iterator: 5
            steps:
                - test_item:
                    name: test_2
                - test_item:
                    name: test_3
```

4.1 Configuration files

A configuration file can be specified in the *.tum* file or by the command line. This configuration file is optional and must be a YAML file.

The configuration files must have the *.yaml* or *.yml* file name extension.

During the test script loading process, the values defined in these configuration files are added to the global variables and are then accessible from the test items and scripts (cf. *global variables*).

The parameter file can be specified in the *.tum* file root:

[6] https://yaml.org/

Listing 4.2: configuration files definition in the main *.tum* test file

```
config_file:
    - config1.yaml
    - config2.yaml

main:
    name: Test example
    [...]
```

Listing 4.3: example of configuration file: param.yaml

```
parameter1: value1
parameter2: 1234
parameter3: <| 12.34 * 2 |>
parameter4:
    - $(parameter1)
    - $(parameter3)
parameter5:
    sub_param1: sub_value1
    sub_param2: $(parameter4)
```

If nothing is specified, the `param.yaml` is automatically loaded, if present in the test directory.

4.1.1 Files loading

The configuration file variables are added to the global variables at load time and stored as declared: `$( )` and `<| |>` in their values are resolved when the value is used. In the example above, `parameter4` follows a later redefinition of `parameter1`. The variables are accessible from TUM test files and from *python* and *lua* function test items.

The resolution rules are detailed in *when values are resolved*.

4.2 Global variables

The global variables feature lets test items and test scripts access a shared, global variables database.

The global variables dataset is populated from various sources:

- the *command line*,
- *built-in values*,
- the *configuration files*,
- some test items results,
- the *helper library API*, accessible from python scripts.

These global variables are used for variable expansion in scripts (see *variables expansion*).

Another possible usage of the global variables is to share persistent data between test steps.

A library allowing python functions to access global variables is available from the python scripts. See details in section *helper library*.

Besides the values from the default *param.yaml* or defined configuration files, the global variables also contain

- built-in specific value (see *below*),
- values returned by test items.

4.2.1 Built-in values

The following keys are automatically accessible through the testium library API (see *helper library*)

- `test_directory`: the absolute path of the directory of the main .tum file,
- `test_main_file`: the main .tum file,
- `os`: the name of the platform which is used. Can be Linux or Windows,
- `host_name`: The name of the host on which testium is running,
- `home`: home directory of the current user,
- `testrun_date`: The date when the test has started (as a string) in format YYYY-MM-DD,
- `testrun_time`: The time when the test has started (as a string) in format HH:MM:SS,
- `test_name`: The name of the file being executed without extension,
- `home`: the path of the current user's home directory,
- `test_outputs`: list of the paths of the test log and test report (if any),
- `last_step_result`: test result of the last step (see *Items common attributes*),
- `ts_start_<item_name>`: timestamp at the start of test item execution (see *Items common attributes*),
- `ts_end_<item_name>`: timestamp at the end of test item execution (see *Items common attributes*),
- `ts_duration_<item_name>`: duration of test item execution in seconds (see *Items common attributes*),
- `cn_<test_name>`: console test item result (see section *console test item*),
- `pfn_<func_name>`: py_func test item result (see section *py_func test item*),
- `lfn_<func_name>`: lua_func test item result (see section *lua_func test item*),
- `cs_<test_name>`: dialog_choices test item result (see section *dialog_choices test item*),
- `loop_param`: loop iterator (available from within a loop item, see *loop test items*),
- `loop_index`: loop index (available from within a loop item, see *loop test items*),
- `loop_count`: loop number (available from within a loop item, see *loop test items*). If the loop is infinite, its value is the Python constant `inf`.

4.2.2 Debug mode

The `test_debug` global variable enables the verbose debug output. It can be set from the Debug button of the Run toolbar, a configuration file (`test_debug: True`) or the command line (`-d test_debug`). See *Debugging your tests*.

4.2.3 Test items entries

All test items attributes can be global variable entries; when a value is written as `$(<name>)`, the name is looked up in the global variables dataset.

4.2.4 References

If the `dialog_references` test item has been included (see *dialog_reference test item*), the global dict will contain the result of this test item in the key `tested_items`.

4.2.5 Dialog values

All dialog returned values are inserted in the global variables entries with the key value being the test item name attribute (see *dialog_value test item*).

4.2.6 Parameter passing, variable expansion and evaluation

One rule governs every attribute value:

- `$( )` substitutes a variable, everywhere.
- `<| |>` evaluates a Python expression, everywhere, and is the only thing that evaluates.
- Without markers, a value is a literal (after substitution).
- A boolean attribute (`condition`, `skipped`, `no_fail`, ...) takes a YAML boolean or a `<| |>` expression producing one; any other value fails the item with an explicit message.

One of the most useful functionalities for scalability and flexibility of the .tum files is the ability to expand variables at test runtime.

It is done by replacing any occurrence of `$(my_global)` with the content of the variable in the global variables entries (see *global variables*).

The variable substitution is recursive and checks all the occurrences of the `$(x)` pattern in a string.

It is also possible to evaluate embedded Python expressions when parameters are passed. It is done by using the `<| expr |>` pattern in a string. *expr* must be a valid Python expression. The modules `os`, `re`, `sys`, `time`, `math`, `json`, `random` and `platform` are available in the expression.

To search a multiline value with a regular expression:

```
- check:
    name: Version line present
    values:
        - <| re.search(r"version:\s+\d+\.\d+", r'''$(out)''', re.M) is not␣
↪None |>
```

- `r'''$(var)'''` keeps the line breaks of the substituted value.
- Do not put YAML quotes around the line. Backslashes (`\d`, `\s`) then pass through unchanged and the pattern is written as in Python.
- `re.M` makes `^` and `$` match at each line.

A variable holding a Windows path needs the same `r` prefix. The substitution is textual, so `"$(test_path)"` becomes `"C:\Users\..."` and Python reads `\U` as an escape sequence:

```
- check:
    name: Path contains Lib
    values:
        - <| "Lib" in r"$(test_path)" |>
```

Below are illustrated simple and more complicated cases of expansion and evaluation depending on their pattern.

Listing 4.4: variables expansion and evaluation

```
- let:
    name: Dynamic variables expansion
    key: $(test)_PASS
    values:
        - expanse_select: <|"$(expanse_select)".replace("o", "a")|>
        - expanse_index: $(expanse_index_$(expanse_select))
        - expanse_table: $(expanse_table_$(expanse_select))
        - expanse_eval: <|$(expanse_index) == 1|>
```

A `let` evaluates its values when the item runs and stores the results. Each pair sees the pairs written above it, as in the example.

4.2.7 When values are resolved

Listing 4.5: param.yaml

```
base_port: 5699
url: localhost:$(base_port)
```

Listing 4.6: test steps

```
- let: {name: Change the port, values: [base_port: 6000]}
- console:
    name: c
    command: ping $(url)     # url is "localhost:6000" here
```

- Item attributes and configuration file values are resolved when they are used, on every execution.
- A redefinition (`let`, `-d`, *API*) applies to every later use, including values derived from the redefined variable, as `url` above.
- A `let` value is the exception: it is evaluated when the `let` runs and the result is stored. It does not change when its source variables change later.
- A variable set to `null` is defined: `$(x)` expands to None.
- Resolved at load time: item `name`, `doc` and `skipped`, the `unittest`/`pytest` `test_file` and `test_method`, file paths (configuration files, includes) and the *jinja templates*.

4.3 Test Items

All *testium* steps are described in sequence as test items in the step list of the main test item (and possibly of a loop test item).

TUM file `main` item is itself a variant of test item with `name` and step-list attributes.

Note

Each test item declares the parameters it accepts. When a `.tum` file uses a key the item does not know, *testium* emits a warning listing the accepted parameter names (catching typos such as `param_filee` for `param_file`); a missing required parameter

aborts loading with an error pointing at the source `.tum` file. Valid existing tests are unaffected.

4.3.1 Items common attributes

All test item types share the common attributes listed in the following table. They are all optional; the table also gives their default values.

Table 4.1: test items common attributes

Parameter name	Default value	Description
`name`	test item type	This is the test item name as displayed in the test tree window of the testium. This attribute is also supported by actions of `console`, `jsonrpc` or `plot` test items. Default value, if not provided, is the test item type.
`stop_on_failure`	`False`	If `stop_on_failure` is set to `True`, the test sequence execution stops on test item failure and no further test items are executed, except those with the `execute_on_stop` attribute set (see below) Not all test item types honor this attribute. For example it makes sense to use it for `unittest` test type because it can contain many sub-tests, but not for sleep test type. In cycles, it means that the child sequence execution is stopped at first failure. It also means that the remaining loops are not executed.
`execute_on_stop`	`False`	When this attribute is set to True, the test item is always run, even on test failure of any test before. This feature is useful, to end the test sequence properly on test failure (switch off power supplies, climatic chamber temperature set to ambient temperature....)
`skipped`	`False`	The test item execution is to be skipped during test sequence execution if set to `True`. It will be displayed as failed in the report.
`no_fail`	`False`	The result of the test step is forced to PASS if this attribute is set to `true`.
`doc`	`""`	Documentation for the test item that appears in the test doc field and the contextual text window in the testium GUI.
`Key`	/	This attribute defines a key which will be attached to the test result and which allows results to be filtered during the report generation.
`report`	/	This attribute defines values (a dictionary) which will be added in the `data` field of the report.
`condition`	/	The test item is executed if its `condition` attribute content is a boolean `True`. see *Conditional execution*.
`process_result`	/	Process an evaluation of the process_result and store it in the result see *Process result* for details.
`expected_result`	/	Expected result value or string. see *Expected result* for details.
`store_result`	/	Store the test result in a global variable. see *Store result* for details.

last test result

The global variable `last_step_result` is automatically set at the end of a test item execution.

If the corresponding test item does not return any actual value, the content of the `last_step_result` variable will be the test success (`PASS`, `FAIL` or `SKIP`).

If the test item returns a value, the `last_step_result` variable will contain the returned value.

The main test items returning a value are:

- *console* test item,
- *jsonrpc* test item,
- *dialog references* test item,
- *dialog value* test item.

Test timings

After the execution of a test step, the following global variables are set :

- `ts_start_<item_name>`
- `ts_end_<item_name>`

and

- duration: `ts_duration_<item_name>`

See *global variables* for more detail on how to access to global variables from test items and scripts.

Skipped test items

A variable named `skipped_test_item` can be defined in the global variable entries or in configuration file (see *config files*) as a list of item to be skipped.

Conditional execution

The `condition` attribute content must be a boolean value or a `<|  |>` expression producing one. Any other value fails the item with an explicit message.

Process result

The `process_result` attribute can be applied to all the test items. However, its behavior differs depending on whether the test item returns a value.

The `process_result` attribute content is a `<|  |>` expression containing the special `$(result)` variable, replaced with the test result value:

```
process_result: <| $(result)['temperature'] |>
```

`process_result` is evaluated before `expected_result`.

If the result of the evaluation is a boolean, the test will be *PASSED* if `True`, and *FAIL* otherwise.

Expected result

The `expected_result` attribute can be applied to all the test items. However, its behavior differs depending on whether the test item returns a value.

The test items returning a value are:

- *dialog_references test item*
- *dialog_value test items*
- *py_func test item*
- *dialog_choices test item*
- *json_rpc test item*

For test items which don't return a value, the `expected_result` attribute content is compared to `PASS` or `FAIL`.

The `expected_result` attribute content is compared literally to the result value; YAML types are preserved (`10` is an integer). A `<| |>` expression is evaluated first:

```
expected_result: 25.0
expected_result: <| ($(reference_table))[$(loop_index)] |>
```

The test is *PASS* if the result equals `expected_result`, and *FAIL* otherwise.

Store result

The `store_result` attribute stores the test result into a named global variable, making it available to subsequent test items via `$(variable_name)`. Like a `let`, it stores a concrete value, evaluated when the item runs.

If the test item returns a value (e.g. `py_func`, `json_rpc`), that value is stored. If `process_result` is also specified, the stored value is the post-processed result.

If the test item produces no value (result is `None`), the stored value is the test status string: `"PASS"` or `"FAIL"`, evaluated after `expected_result` but before `no_fail`. This ensures the real outcome is captured even when `no_fail: True` would otherwise mask a failure.

Listing 4.7: Store a function return value

```
- py_func:
    name: Read sensor
    func_name: read_temperature
    store_result: temperature

- py_func:
    name: Check temperature in range
    func_name: check_range
    param: [$(temperature), 20, 30]
```

Listing 4.8: Store a post-processed value

```
- py_func:
    name: Get firmware version string
    func_name: get_version
    process_result: "<| '$(result)'.split('.')[0] |>"
    store_result: fw_major
```

Listing 4.9: Store the pass/fail status of a test with no return value

```
- console:
    name: Send command
    console_name: device
    steps:
        - writeln: reboot
        - read_until: {expected: "ready", timeout: 10}
    store_result: reboot_status

- py_func:
    name: Use reboot status
    func_name: log_status
    param: [$(reboot_status)]
```

key and report attributes

Listing 4.10: Example of key and report common attributes usage

```
- check:
    name: Example of result specific to the step 001
    values:
        - <| "$(last_step_result)" == "PASS" |>
    key:
        - GID-1510554_step_1
    report:
        reported_list: <| random.sample(range(0,20), k=10) |>
        reported_float: <| math.sqrt(float(1)) |>
        reported_str: This is my reported sentence
```

4.3.2 Container items GUI default folding

The container items are items which are the parent of other test items. For example loops and groups are container test items.

In the GUI, to make a container test item folded when the test is opened, place a `.` character before the test item declaration.

See an example below:

Listing 4.11: example of `loop` folded by default in the GUI

```
- .loop:
    doc: An example loop
    name: An example loop
...
```

4.3.3 check test item

The check test item returns the result of a python string evaluation:

Listing 4.12: example of check test item usage

```
- check:
    name: check test item example
    values:
        - <| "tictactoe" in "$(my_global_string)" |>
```

4.3.4 console test item

The console test item is of the form:

Listing 4.13: example of console test item usage

```
- console:
    name: test name in GUI
    console_name: console name in dict
    steps:
        - open:
            protocol: telnet
            telnet_host: $(target_ip)
            telnet_port: $(target_port)
        - writeln: reset
        - read_until: {expected: U-Boot, timeout: 50}
        - write: $(boot_vxworks_1)
        - writeln: $(boot_vxworks_2)
        - read_until:
            expected: U-Boot
            timeout: 15
        - read_until:
            expected: Something that will never occur
            timeout: 5
            no_fail: True
            mute: True
        - close:
```

Attributes

Besides the common test item attributes, the console test item has specific attributes:

- `console_name`: console instance name
- `write_delay`: optional parameter giving the delay to wait in milliseconds between each character sent.
- `steps`: a sequence of actions to be applied to the console, as listed above.

The console test item steps accept the parameters and configurations defined in the next sections.

All the following actions support the `name` attribute. The `name` is concatenated with the step type in the *testium* GUI, and repeated in the test log and reports.

open action

The open action initializes the console with the attributes defined as described below. The console instance is then added to the `console_instances` entry of the global variables (cf *global variables*).

Open step accepts the following attribute:

- `protocol`: Setting of the console protocol; supported protocols are listed in the table below
- The other attributes depend on the protocol in use and are listed in the table below. The protocol and connection parameters are resolved on every execution of the open action.

Table 4.2: console protocols

Protocol	protocol parameter	Description
`telnet`	`telnet_host`	hostname of the target.
	`telnet_port`	port of the telnet server of the target.
`ssh`	`ssh_host`	Hostname or IP address of the target.
	`ssh_user`	User name for the SSH connection.
	`ssh_pwd`	Password (optional).
`serial`	`serial_port`	Serial port to the target.
	`serial_baudrate`	Baud rate of the serial connection.
	`buffered`	Optional boolean parameter. If `False`, it forces the console to read the device directly. Default: `True`.
`rawtcp`	`tcp_host`	hostname of the target.
	`tcp_port`	port of the rawtcp server of the target.
`terminal`	`terminal_path`	Path of the terminal console.
	`shell`	Shell to execute in the terminal Default: /usr/bin/env bash

- `log`: is available only for Telnet and Serial console and is a path to a folder or a file, where the log will be stored.
- `newline`: line ending appended by `writeln`: `lf` (default), `crlf` or `cr`. Use `crlf` or `cr` for serial devices and network equipment expecting carriage returns.
- `dialect`: shell dialect used by the exec action: `sh`, `cmd`, `powershell` or `none`. By default it is guessed from the shell for the `terminal` and `ssh` protocols, and `none` for the device protocols (telnet, rawtcp, serial).

`close` action

The `close` action closes the console devices and removes its instance from the `console_instances` list accessible in the global variables (cf *global variables*).

No parameters required for this action.

`write` action

`write` action takes as parameter the string to be written on the console.

`writeln` action

writeln function is similar to the write function except that a 'n' (newline) character is sent at the end of the string to be written.

`exec` action

The exec action sends a command to a shell console and waits until the command is finished. *testium* appends a unique per-call marker to the command line; seeing the marker in the output means the command completed. This works identically for a local shell, an SSH session or a shell started inside another shell. No prompt configuration is needed.

Listing 4.14: exec: short and detailed forms

```
- exec: make all

- exec:
    cmd: ./deploy.sh
    timeout: 30
    process_result: "<| 'done' in r'''$(result)''' |>"
```

Parameters:

- `cmd`: the command line to run (single line). The short form `- exec: <command>` is equivalent to `- exec: {cmd: <command>}`.
- `timeout`: seconds before giving up (negative or absent: infinite). An interactive command that never returns to the shell (an editor, a password prompt) never prints the marker; the timeout is the safety net.
- `mute`: if `True`, the exchanged data is not logged.

The command output (marker removed) is stored in the item result and in the global variable `cn_<test_name>`, like `read_until`.

The console must have a shell `dialect` (see the open action). On a `none` console (network equipment, an application managing its own prompt), `exec` fails with an explicit message: synchronize those with `writeln` and `read_until` instead. A shell started inside the session that speaks another dialect (PowerShell launched from bash) is not handled by `exec`.

The marker travels in the console stream, so it is visible in the console logs; use `mute` to hide the exchange.

`read_until` action

The `read_until` action waits for a string pattern from the console. Its parameters are listed below:

- `expected`: the value to wait for, matched literally (never evaluated as an expression). A list is also accepted; the action succeeds when any entry of the list is seen.
- `regex`: Boolean value (default `False`). When `True`, each `expected` entry is a Python regular expression searched in the received text.
- `timeout`: Timeout setting for the action (in seconds)
- `no_fail`: Boolean value (`True` or `False`). If `True`, no error is reported when the expected input is not read
- `mute`: Boolean value (`True` or `False`). If `True`, the data read is not logged

Listing 4.15: matching several values, and with a regular expression

```
# succeeds as soon as one of the three strings is received
- read_until:
    expected: [login:, "Password:", "$ "]
    timeout: 10

# regex: wait for "version X.Y.Z" with any numbers
- read_until:
    expected: 'version \d+\.\d+\.\d+'
    regex: True
    timeout: 5
```

The text read by the `read_until` action is stored in the global variable named `cn_<test_name>` (See *global variables* for more detail on accessing global variables from test items and scripts). When a list of values is given, the report also records, under the `matched` key, which pattern actually matched.

Note

With `regex: True`, only the last 64 KiB of received text are searched. Older text is not searched again.

In the example above, the global variable `$(cn_test name in GUI)` would be created at the end of the step. It contains the data that was read.

4.3.5 dialog_choices test item

This test item displays a dialog asking a question and waits for the user to select entries from a defined list of items.

These selectable items can be passed as a tree. Two controls above the tree select or deselect every entry and fold or unfold the tree, as in the main window.

The Figure 4.1 displays an example of this item.

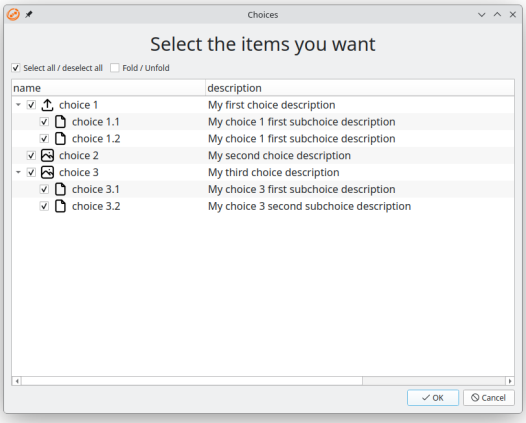


Fig. 4.1: choices dialog

The item parameters corresponding to Figure 4.1 are shown below.

Listing 4.16: example of `choices_dialog` test item usage

```
- dialog_choices:
    name: Choices
    question: Select the items you want
    icon: $(test_directory)/document.png
    choices:
      - name: choice 1
        description: My first choice description
        icon: $(test_directory)/document-save.png
        choices:
          - name: choice 1.1
            description: My choice 1 first subchoice description
            icon: $(test_directory)/Label.png
```

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```
        - name: choice 1.2
          description: My choice 1 first subchoice description

    - name: choice 2
      description: My second choice description
      icon: $(test_directory)/image.png

    - name: choice 3
      description: My third choice description
      icon: $(test_directory)/image.png
      choices:
        - name: choice 3.1
          description: My choice 3 first subchoice description

        - name: choice 3.2
          description: My choice 3 second subchoice description
          icon: $(test_directory)/Label.png
```

Attributes

The supported attributes of the `dialog_choices` test item are:

- `question`: Question to be displayed in the dialog box.
- `choices`: List of the choices presented to the user.
- `icon`: Optional. Path of the icon used in the selection tree, for all the items by default.
- `auto_result`: Optional. Selection used in batch / non-interactive mode (choice path or label). If not set, the step fails.

Choices attribute content

Each choice element is a dictionary which can have the following attributes

- `name`: name of the choice.
- `description`: description of the choice.
- `icon`: Optional. Path of the icon displayed in the selection tree in front of the corresponding choice.
- `choices`: List of sub-choices presented to the user (recursive).

Feature

The `dialog_choices` test item creates the `cs_<name of test item>` entry in the global dictionary.

In the example above, the global variable name containing the result of the test item would be `cs_Choices`, and it would contain an object of this form:

Listing 4.17: example of result of the `dialog_choices` test item

```
[
 {'name': 'choice 1',
  'checked': True,
  'choices': [ {'name': 'choice 1.1', 'checked': True},
```

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```
                {'name': 'choice 1.2', 'checked': False} ]
 },
 { 'name': 'choice 2',
   'checked': False},
 { 'name': 'choice 3',
   'checked': True,
   'choices': [ {'name': 'choice 3.1', 'checked': False},
                {'name': 'choice 3.2', 'checked': True} ]
 }
]
```

See *global variables* for more detail on how to access global variables from test items and scripts.

4.3.6 dialog_image test item

This test item displays an image within a dialog box.

`dialog_image` test item has the following description format

Listing 4.18: example of `dialog_image` test item usage

```
- dialog_image:
    name: dialog image test item
    question: operator question
    filename: imageToBeDisplayed.jpg
```

Attributes

`dialog_image` has the following specific attributes:

- `question`: Question to be displayed in the dialog box
- `filename`: File name of the image to be displayed in the dialog box.
- `auto_result`: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails.

Feature

The test returns a FAIL if the answer is No and PASS if yes.

4.3.7 dialog_message test item

This test item displays a simple dialog showing a message. `dialog_message` test item has the following description format

Listing 4.19: example of dialog_message test item usage

```
- dialog_message:
    name: dialog value test item
    question: operator question
```

Attributes

dialog_message has the following specific attribute:

- question: Sentence to be displayed in the dialog box
- auto_result: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails.

Feature

Displays the message; no value is returned.

4.3.8 dialog_note test items

This test item displays a simple dialog that lets the operator enter text, and prints the entered text in the logs.

dialog_note test item has the following description format

Listing 4.20: example of dialog_note test item usage

```
- dialog_note:
    name: dialog value test item
    question: operator question
```

Attributes

dialog_note has the following specific attribute:

- question: Question to be displayed in the dialog box
- auto_result: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails; cancel marks it cancelled; any other value makes it succeed with auto_value.
- auto_value: Optional. Note text used in batch mode when auto_result is set.

Feature

Prints the entered text in the log.

4.3.9 dialog_question test item

This test item displays a dialog asking a yes/no question; the answer determines the test result.

dialog_question test item has the following description format

Listing 4.21: example of `dialog_question` test item usage

```
- dialog_question:
    name: dialog value test item
    question: operator question
```

Attributes

`dialog_question` has the following specific attribute:

- `question`: Question to be asked in the dialog box
- `auto_result`: Optional. Answer used in batch / non-interactive mode ('yes' or 'no'). If not set, the step fails.

Feature

The test returns a FAIL if the answer is No and PASS if yes.

4.3.10 dialog_references test item

This test item displays a dialog asking a question and waiting for references of the devices under test.

This test item has the following format:

Listing 4.22: example of `dialog_references` test item usage

```
- dialog_references:
    name: ask for a reference
    question: Please give the reference of the product
    reference:
        - ref 1
        - ref 2/rev
```

The example above displays the dialog box in Figure 4.2.

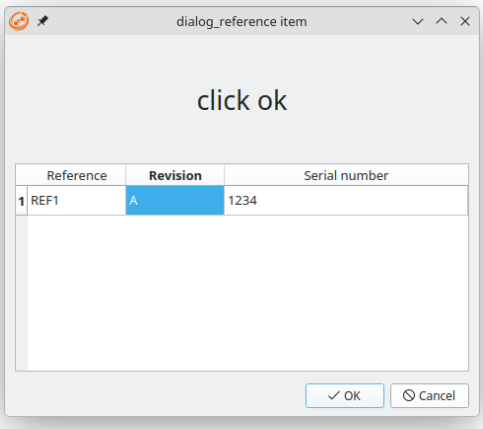


Fig. 4.2: dialog reference

Attributes

`question` and `reference` are mandatory.

- `question`: Question to be displayed in the dialog box.
- `reference`: Each entry in this list corresponds to one row to fill in the dialog.
- `auto_result`: Optional. Outcome used in batch / non-interactive mode. If set, the step succeeds with the pre-filled references; if not set, it fails.

Every field for a reference can be pre-filled by separating each field with a '/' (cf Figure 4.2).

Feature

The dialog references test item creates the `tested_items` entry in the global_dict global variable. This entry is a list of dictionaries of this form:

Listing 4.23: example of `tested_items` global variable result of `dialog_reference` test item

```
[{'reference': 'XXXXX', 'revision': 'YYYYY', 'serial': 'ZZZZZ'}, …]
```

4.3.11 dialog_value test items

This test item displays a simple dialog asking a question and returning the entered value.

`dialog_value` test item has the following description format

Listing 4.24: example of `dialog_value` test item usage

```
- dialog_value:
    name: dialog value test item
    question: operator question
```

Attributes

`dialog_value` has the following specific attribute:

- `question`: Question to be displayed in the dialog box
- `default`: default value to place in the dialog form (optional)
- `auto_result`: Optional. Outcome used in batch / non-interactive mode. If not set, the step fails; `cancel` marks it cancelled; any other value makes it succeed with `auto_value`.
- `auto_value`: Optional. Value used in batch mode when `auto_result` is set.

Feature

The entered value is stored in the global variables under the `dialog_value` test item name as key.

4.3.12 git test item

The `git` test item records the state of one or more git repositories. It has the following description format

Listing 4.25: `git` test item usage example

```
- git:
    name: git test item
    repo: [$(test_directory), "/path_to/another/repo"]
```

Attributes

- repo: a path, or list of paths, to the root of the git repository or repositories to track.

4.3.13 group test item

This element is of the following form:

Listing 4.26: `group` test item usage example

```
- group:
    name: Group Item
    condition: <| "$(OS)" == "Linux" |>
    steps:
        - unittest:
            test_file: test_prod_alpha_13.py
            test_method:
                ...
        - sleep:
            timeout: 10
```

The `group` element is used to run a sequence of items as a group.

Attributes

- The `steps` list describes the sequence executed in the group. It is a list of any *testium* test items.

4.3.14 jsonrpc test item

The *jsonrpc* test item is used to access jsonrpc servers, by sending queries and analysing the answers. It supports JSONRPC v1.0[7] or v2.0[8].

This test item can access the jsonrpc server by using an existing *console* or directly using a UDP protocol. Two low level *adapters* can then be chosen: `udp` or `console`.

Example of `jsonrpc` test item with the console adapter:

Listing 4.27: `json_rpc` test item usage example

```
- json_rpc:
    name: JSONRPC console Query
    doc: JSONRPC console Query not waiting (only send)
    console:
      name : jsonrpc_server
      prompt: "@@>"
    timeout: 1
    version: "2.0"
    steps:
```

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[7] https://www.jsonrpc.org/specification_v1
[8] https://www.jsonrpc.org/specification

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```
        - query:
            method: echo
            params:
                a: Hello world
                b: [0, 1, 2, 3]
            id: 3095372
            no_wait: True

- [...]

- json_rpc:
    name: JSONRPC console Reception
    doc: JSONRPC console reception of the previous request
    console: {name : jsonrpc_server}
    timeout: 1
    steps:
        - receive:
            name: console reception
            id: 3095372
            timeout: 0.5
```

Attributes

the jsonrpc attributes are:

- `timeout`: global communication timeout in seconds. It is a floating point number.
- `version`: "1.0" or "2.0" (as a string) depending on the version of the JSONRPC standard which is supported.
- `mute`: a boolean giving the verbosity of the jsonrpc exchanges on the log output.
- An *Adapter* is to be chosen between:
  - Console,
  - UDP,
- `steps`: a sequence of actions as described in the sections below.

Steps

A jsonrpc step can be one of the following:

- `open`: used by the UDP adapter to open the socket explicitly,
- `close`: used by the UDP adapter to close the socket explicitly,
- `query`: performs a complete or partial JSONRPC call,
- `receive`: used to receive the JSONRPC result of call previously done by the `query` action.

If no `expected_value` attribute is defined for `query` or `receive` actions, the success of the step will depend on the value returned by the JSONRPC frame. The protocol itself defines a way to report whether the remote procedure succeeded or failed.

All the actions support the `name` attribute. The `name` is concatenated with the action type in the *testium* GUI, and repeated in the test log and reports.

adapter attributes

The adapters attributes are listed in the table below.

Table 4.3: jsonrpc adapters

adapter	attribute attribute	*type*	Description
Console	`console`	*dictionary*	The console adapter configuration
	`console.name`	*string*	The name of the console which will be retrieved from the *global variables*. See also the *console test item*.
	`console.prompt`	*string*	optional suffix that terminates the jsonrpc frame.
UDP	`udp`	*dictionary*	The UDP adapter configuration
	`udp.server`	*string*	UDP server hostname or IP address. A multicast address (224.0.0.0/4) enables the multicast mode (see below).
	`udp.snd_port`	*integer*	UDP server listening port
	`udp.rcv_port`	*integer*	UDP answer reception port (on client side)
	`udp.bufsize`	*integer*	the maximum expected size of the buffer received while waiting for a jsonrpc frame.
	`udp.multicast_if`	*string*	multicast mode only: IP address of the local interface used to send to and join the group (default: the kernel default interface).

Multicast

When `udp.server` is a multicast group address (224.0.0.0/4), the open action sends the requests to the group and joins it on the reception socket.

The response scheme is decided by the *server*, not by *testium*; no client configuration is needed, both are received transparently on `udp.rcv_port`:

- the server answers in unicast to the source address of the request (the usual case): received because the socket is bound to `rcv_port`,
- the server answers to the multicast group itself (discovery-style protocols): received because open joined the group. The server must then address the group on the client reception port.

Use `udp.multicast_if` to pin the exchanges to a given local interface (e.g. `127.0.0.1` for a same-host test bench); otherwise the kernel routing decides which interface carries the multicast traffic; on a multi-homed host the request may leave through the wrong network.

Warning

The UDP socket is created once by the open action and shared, through `rcv_port`, by every later `query`/`receive` item. `multicast_if` only acts at socket creation: set it on the item holding the open action. On any other item it is ignored (a warning is logged).

`open` action

The `open` jsonrpc action is only used with the *UDP adapter* but is mandatory before any `query` action.

No parameter is required.

`close` action

The `close` jsonrpc action is only used with the *UDP adapter* but is mandatory after JSONRPC transfers are finished.

No parameter is required.

`query` action

The `query` jsonrpc action has the following attributes:

- `method`: JSONRPC method to be called,
- `params`: JSONRPC param (must conform to the version defined above), by default it is an empty list.
- `id`: JSONRPC id. If not defined or starts with `rand`, it is chosen randomly. Otherwise it must be an integer value,
- `timeout`: reception timeout in seconds. It is a floating point number. It is by default the jsonrpc timeout.
- `no_wait`: Optional boolean. False by default. This attribute defines if the reception is performed in this step (reception can be done separately, in the `receive` action described below),

`receive` action

The `receive` jsonrpc action has the following attributes:

- `id`: JSONRPC id as an integer value,
- `timeout`: reception timeout in seconds. It is a floating point number, It is by default the jsonrpc timeout.

4.3.15 let test item

This element is of the following form:

Listing 4.28: `let` test item usage example

```
- let:
    name: Let Item
    values:
        - key1: value1
        - key2: value2
        - key3: <| $(variable)[$(loop_index)] |>
```

The `let` element is used to set values in the global dictionary.

The values are evaluated when the item runs and the results are stored: a value derived from another variable does not change when that variable changes later. Each pair sees the pairs written above it. Configuration file values behave differently: they are *resolved when used*.

Attributes

- The `values` list gives the {<key>, <value>} pairs to set in the global dictionary,
- `condition`: the item runs only when the expression is true (*common attribute*),
- key: expected-result accounting key (*common attribute*).

4.3.16 loop test items

This element is of the following form:

Listing 4.29: `loop` test item usage example

```
- loop:
    name: Cycle Temperature
    iterator: 10
    steps:
        - unittest:
            test_file: test_prod_rio6_8093.py
        - py_func:
            name: function test item
            file: scriptTestFile.py
            func_name: funcToBeExecuted
            param:
                - $(loop_param)
    exit_condition:
        file: script_name.py
        func_name: methodName
```

The loop element executes repeatedly the `steps` sequence of items.

Iteration is controlled by the `iterator` attribute, described later in this chapter.

Attributes

Below are described loop test item specific attributes.

- `Iterator`: giving the number of loop iteration (see dedicated chapter below).
- `steps`: describes the sequence executed at each cycle; it is a list of any of the testium test items.
- `exit_condition`: defines when to exit the loop. The loop stops when the condition evaluates to True. exit_condition attributes are:
  - `time`: the loop stops after the time (in minutes) is elapsed (optional)
  - `value`: the loop stops when the content of the value attribute is True (optional)
  - `file`: the script file that contains a function executed at each iteration. Only python script format is supported (optional if another exit_condition attribute is defined)
  - `func_name`: the function to execute on each loop when the file attribute is defined. The function referenced by the `func_name` attribute must have two parameters:

the current loop iterator value and the report, even if they are not used. This attribute is mandatory if the file attribute is defined.

Listing 4.30: `loop` exit condition

```
- loop:
    ...
    exit_condition:
        file: script_name.py
        func_name: methodName
```

Iterator

The iterator attribute can be of the following types:

- An integer giving the number of iterations,
- A list. The number of elements of the list gives the loop number, and the list member are the consecutive loop parameters,
- Undefined. Then cycle loops until the exit condition is reached.

Loop variables

The following loop variables are automatically defined:

- `$(loop_param)`: parameter of the loop. It contains the iterator value.
- `$(loop_index)`: index of the loop, starting with 0 and incremented at each cycle.
- `$(loop_index_inverse)`: index counting down, starting from the iteration count minus 1 and decremented at each cycle.
- `$(loop_count)`: loop total iteration number. If the number of loops is undefined its value is the python `inf`.

When these variables are found in a parameter, an attribute, etc, testium searches the test hierarchy upward for the enclosing loop and substitutes that loop's value.

If more than one loop exists in the test item hierarchy, the lowest level loop iterator is used.

4.3.17 lua_func test item

The `lua_func` test item is used to execute custom lua 5.4 scripts with the given input parameters.

The `lua_func` test item is of the form:

Listing 4.31: `lua_func` Lua function example

```
local module = {}

function module.dummy_func(param1, param2, param3, param4)
    ...
    return 10
end

return module
```

Listing 4.32: corresponding `lua_func` tum extract

```
- lua_func:
    name: lua function test item
    file: script_file.lua
```

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```
    func_name: dummy_func
    param:
        - 123
        - 0.123
        - True
        - $(global_dict_key)
    expected_result: 10
```

Attributes

Besides the common test item attributes, the `lua_func` item has specific attributes; `file` and `func_name` are mandatory. Both are resolved on every execution: in a loop, a `$( )` value can select a different function per iteration.

- `file`: the script file name that contains the function to be executed. Only Lua script format is supported.
- `func_name`: The function name to be executed.
- `param`: This is a list of parameters that are passed to the function in the order they are presented in the script. These parameters are not mandatory and depend on the function prototype.
- `context_id`: Optional. When set, all `lua_func` items sharing the same `context_id` value run inside the same persistent Lua subprocess for the duration of the test. See *lua_func context* for details.

Listing 4.33: `lua_func` test item example of usage

```
- lua_func:
    name: activity
    file: script_name.lua
    func_name: methodName
    param:
        - $(my_param)
```

The result of the function (after optional post-processing) is stored in the global variable named `lfn_<item_name>` (See *global variables* for more detail on how to access to global variables from test items and scripts).

In the example above, the global variable `$(lfn_activity)` would be created at the end of the item execution. It would contain the resulting value of the methodName Lua function.

The `lua_func` result is always PASS, unless the called function raises an error or the `expected_result` attribute is used.

If the called function returns several values, they are kept as a list in the result; a single return value stays a scalar.

Sharing state between `lua_func` calls

Each `lua_func` item without a `context_id` runs in a dedicated subprocess that is started and stopped around the call. Module-level variables are not preserved between two such items.

Inside a `lua_func` script, the `tm` module exposes `tm.setgd` and `tm.gd` to read and write the testium global dictionary of the test process. Values stored this way are accessible from any subsequent test item without requiring a shared subprocess.

Listing 4.34: sharing a value via the global dictionary

```lua
local tm = require("tm")
local module = {}

function module.produce(val)
    tm.setgd("my_shared_value", val)
    return val
end

function module.consume()
    return tm.gd("my_shared_value")
end

return module
```

When `context_id` is set, all `lua_func` items that share the same identifier reuse the same persistent subprocess. This allows Lua-side state (upvalues, module cache) to be retained across calls beyond what `tm.setgd` persists.

Listing 4.35: `lua_func` items sharing a persistent subprocess

```yaml
- lua_func:
    name: produce value
    file: my_script.lua
    func_name: produce
    context_id: my_context
    param:
        - hello

- lua_func:
    name: consume value
    file: my_script.lua
    func_name: consume
    context_id: my_context
    expected_result: hello
```

The shared subprocess is automatically stopped at the end of the test run.

Lua Interpreter environment setup

Some global variables have an impact on the `lua_func` test item behavior:

- `lua_bin`: This optional global variable can be used to define the lua executable path. If not defined, the lua interpreter is looked up in the default system locations (PATH).
- `lua_env`: This global variable can be used to define environment variables for the lua script execution environment. Only *PATH*, *LUA_PATH*, and *LUA_CPATH* are supported.

Listing 4.36: example of configuration file: param.yaml

```
[...]
lua_env:
  PATH: "/my/path/"
  LUA_PATH: "/my/lua/modules/?.lua;;"
  LUA_CPATH: "/my/lua/modules/?.so;;"
[...]
```

4.3.18 parallel test item

This element is of the following form:

Listing 4.37: `parallel` test item usage example

```
- parallel:
    name: My parallel block
    sync: all
    branches:
      - name: Branch A
        steps:
            - py_func:
                name: Long operation
                file: long_op.py
                func_name: do_work
      - name: Branch B
        wait_for:
            condition: <| "$(ready_flag)" == "True" |>
            timeout: 30
        steps:
            - let:
                name: Mark done
                values:
                    - branch_b_done: true
```

The `parallel` element runs several sequences of items concurrently. Each inner sequence is called a *branch* and runs in its own thread. The parent test item waits for branches to finish according to the `sync` policy.

Attributes

- `branches`: required. A list of branches to execute concurrently. Each branch has a `name` and a `steps` list (same structure as a `group` item). It can also declare a `wait_for` precondition (see below).
- `sync`: optional, defaults to `all`.
  - `all`: the parallel item completes when *every* branch has finished. The result is PASS if no branch returned `FAIL` (skipped or disabled branches are ignored, like in `group`); otherwise `FAIL`.
  - any: the parallel item completes as soon as the *first* branch finishes. The remaining branches are stopped (their next test items are not executed). The result is PASS if at least one branch succeeded.
- `no_fail`: optional. When `true`, a `FAIL` result is forced to PASS for the parallel item itself (same semantics as for any test item). Branches keep their own result.

Branch attributes

Each entry of `branches` is a dict with the following attributes:

- `name`: required. The branch name. Used in reports and as a prefix in the live log output (each line printed by the branch is prefixed with the branch name in square brackets, e.g. `[Branch A]`, so concurrent branches stay readable).
- `steps`: required. The list of test items executed sequentially inside the branch.
- `wait_for`: optional. Forces the branch to wait until a condition is met before running its steps. If the timeout elapses, the branch returns `FAIL` (the steps are not run). Sub-attributes:
  - `condition`: a testium expression evaluated repeatedly (every 100 ms) until it returns `True`.
  - `timeout`: maximum wait, in seconds. Defaults to 30.

Reporting

Each branch produces its own row in the SQLite report (with type `Parallel branch`), in addition to the parent `Parallel` row. The `log` column of each row contains only the output emitted from that branch's thread, so logs are never mixed between concurrent branches.

In the live (terminal / GUI) output, lines emitted from a branch are prefixed with the branch name in square brackets (e.g. `[Branch A]`). The prefix is not stored in the SQLite log column.

Notes

- A `sleep` item inside a branch is interruptible: if another branch finishes first under `sync: any`, slow `sleep` items are aborted within ~50 ms.
- A `py_func` or `console` item inside a branch is not interruptible: a `sync: any` stop will only take effect after the current item returns. The branch will then skip its remaining steps.
- When a user disables a branch in the GUI tree, the branch returns SKIP instantly without affecting the others (it does *not* count as the first finished branch under `sync: any`).

4.3.19 plot test item

This test item is used to display runtime values of test variables, or any changing value, in a separate external window.

The plot window is defined using the `plot` test item:

Listing 4.38: `plot` test item usage example

```
- plot:
    name: test name in GUI
    plot_name: plot identifier
    steps:
        - open:
        - add:
            ...
```

Attributes

In addition to common test items attributes, the `plot` test item has specific attributes:

- `plot_name`: plot window instance name.
- `steps`: a sequence of actions to be applied to the plot window. More than one action can be executed in a `plot` item.

The plot test item can accept the actions described in further sections.

All the following actions support the `name` attribute. The `name` is concatenated with the action type in the *testium* GUI, and recalled in the test log and reports.

`open` action

This action initializes and opens the plot window with the corresponding attributes as defined below.

This action accepts one optional `log_path` parameter defining the path where the plot line values are stored in CSV format.

Listing 4.39: `plot` open action

```
- plot:
    name: Open the plot window
    plot_name: plot identifier
    steps:
        - open:
            name: open the plot
            log_path: $(test_directory)/tmp
```

`close` action

The `close` action closes the plot window and removes it from the managed instances of *testium*.

This action does not have mandatory parameters. However, `close` optional action parameters are:

- `wait_dialog_exit`: Boolean value. If set to True, the window is kept open until the user closes it manually.
- `timeout`: Value expressed in seconds. It is active if the `wait_dialog_exit` is set to True. If this parameter is defined, and if not closed manually, the dialog window is kept open until the timeout elapses.

Listing 4.40: `plot close` action

```
- plot:
    name: Closes the plot
    plot_name: plot identifier
    steps:
        - close:
            wait_dialog_exit: True
            timeout: 600
```

Note

When the `close` action is entered, the `periodic` plots are stopped.

add action

The add action is used to add a single data point to the plot window.

Listing 4.41: plot add action

```
- plot:
    name: Add to the plot
    plot_name: plot identifier
    steps:
        - add:
            name: add value 1 & 2
            value1: $(loop_index)
            value2: $(loop_index)+2
```

The parameter of the add action is a dictionary of (*key*, *values*) pairs where the *key* is the plot line name and *value* is the numeric value to add to the plot line.

If the *value* is a string rather than a number, it is evaluated as a python expression.

periodic action

This action calls a python function periodically and uses its result to update the plot.

periodic plots are updated automatically and don't require further steps in a test sequence, once executed.

periodic action parameters are:

- period: period of the automatic value update, in seconds or as a *duration string*. A variable or a <| |> expression is accepted.
- file: python file containing the function to call.
- func_name: the name of the python function to be periodically called.
- eval: optional parameter allowing post-processing of the function result.

The eval parameter of the periodic action is a python evaluable string in which the $(result) keyword is replaced by the actual function call result.

The result of the action must be a dictionary of (*key*, *values*) pairs where the *key* is the plot line name and *value* is the numeric value to add to the plot line.

Listing 4.42: plot periodic action

```
- plot:
    name: Add periodic to the plot
    plot_name: plot identifier
    steps:
        - periodic:
            period:         1
            file:           $(test_path)$(psep)plot.py
            func_name:      random_value
            eval:           '<| {"periodic": $(result)} |>'
```

last_value action

The last_value action stores the last values added to the plot (periodically or not) in the global variables entries.

last_value action parameters are:

- name: Optional parameter giving the list of plot lines (measurements) to return. If it is not defined, all the plot lines are returned.

Listing 4.43: plot last_value action

```
- plot:
    name: Plot measure_1 value
    plot_name: plot identifier
    steps:
        - last_value:
            name: [measure_1]
```

The result of the action is stored in the global variable named plv_<item_name> in the example above, it would be $(plv_Plot measure_1 value). See *global variables* for more detail on how to access to global variables from test items and scripts.

export action

The export action saves the plot window data in various formats to the filesystem.

Listing 4.44: plot export action

```
- plot:
    name: Plot export
    plot_name: plot identifier
    steps:
        - export: $(my_custom_path)/plot_export.pdf
        - export: $(my_custom_path)/plot_export.csv
```

At the time of writing of this documentation, .pdf and .csv files are supported.

4.3.20 py_func test item

The py_func test item is used to execute custom python scripts with the given input parameters.

There are two modes for executing a py_func item. The *class* mode and the *function* mode.

class py_func item

This is the normal way of calling some custom python code.

A class must be defined and derived from FunctionItem from the py_func.tm module.

From this class it is possible to define some custom reported values with the following API

- reportValue(key, value): This FunctionItem method adds a value to the report,
- reportedValues(): This FunctionItem method returns the values reported so far.

Listing 4.45: py_func test item implementation example

```
import py_func.tm as tm

class TestItemPyFunc(tm.FunctionItem):

    def exec(self, param1, param2, param3, param4):
        ...
        self.reportValue('my_reported_value', reported_value)
        print(self.reportedValues())
        return 10
```

The exec method of the FunctionItem derived class is executed while running the py_func test item.

Listing 4.46: class py_func test item usage

```
- py_func:
    name: function test item
    file: scriptTestFile.py
    func_name: TestItemPyFunc
    param:
        - 123
        - 0.123
        - True
        - $(global_dict_key)
    expected_result: 10
```

legacy py_func

The legacy py_func test item is of the form:

Listing 4.47: legacy py_func python function example

```
def dummy_func(param1, param2, param3, param4):
    ...
    return 10
```

There is no possibility to access the report features in that mode.

Listing 4.48: corresponding py_func tum extract

```
- py_func:
    name: function test item
    file: scriptTestFile.py
    func_name: funcToBeExecuted
    param:
        - 123
        - 0.123
        - True
        - $(global_dict_key)
    expected_result: 10
```

Attributes

Besides the common test item attributes, the py_func item has specific attributes; `file` and `func_name` are mandatory. Both are resolved on every execution: in a loop, a `$( )` value can select a different function per iteration.

- `file`: the script file name that contains the function to be executed. Only python script format is supported.
- `func_name`: The function name to be executed.
- `param`: This is a list of parameters that are passed to the function in the order they are presented in the script. These parameters are not mandatory and depend on the function prototype.
- `context_id`: Optional. When set, all py_func items sharing the same `context_id` value run inside the same persistent Python subprocess for the duration of the test. See *py_func context* for details.
- `debug`: Optional. When `true`, the subprocess waits for a debugger to attach before running the function. See *debugging* below.

Listing 4.49: py_func test item example of usage

```
- py_func:
    file: script_name.py
    func_name: methodName
    param:
        - $(my_param)
```

The result of the function (after optional post-processing) is stored in the global variable named `pfn_<item_name>` (See *global variables* for more detail on how to access to global variables from test items and scripts).

In the example above, the global variable `$(pfn_function test item)` would be created at the end of the item execution. It would contain the resulting value of the funcToBeExecuted python function.

The py_func result is always PASS, unless the called function raises an exception or the `expected_result` attribute is used.

Sharing state between py_func calls

Each py_func item without a `context_id` runs in a dedicated subprocess that is started and stopped around the call. State cannot be shared between two such items using module-level variables.

Inside a py_func script, `tm.setgd` and `tm.gd` read and write the testium global dictionary. Values stored this way are accessible from any subsequent test item, including other py_func items, without requiring a shared subprocess.

Listing 4.50: sharing a value via the global dictionary

```
import py_func.tm as tm

def produce(val):
    tm.setgd("my_shared_value", val)
    return val

def consume():
    return tm.gd("my_shared_value", None)
```

When `context_id` is set, all py_func items that share the same identifier reuse the same persistent subprocess. This allows sharing any Python object across calls, including objects that cannot be transmitted to other processes.

Listing 4.51: sharing an object via `context_id`

```
import py_func.tm as tm

def open_connection():
    tm.setgd("conn", MyConnection())
    return "ok"

def use_connection():
    conn = tm.gd("conn")
    return conn.status()
```

Listing 4.52: py_func items sharing a persistent subprocess

```
- py_func:
    name: open connection
    file: my_script.py
    func_name: open_connection
    context_id: my_context
    expected_result: ok

- py_func:
    name: use connection
    file: my_script.py
    func_name: use_connection
    context_id: my_context
    expected_result: opened
```

The shared subprocess is automatically stopped at the end of the test run.

Debugging your functions

Setting `debug: true` on a `py_func` item lets you debug the function step by step from your IDE. A step-by-step tutorial is provided in `doc/debug_tutorial.md` of the source repository; this section is the reference. Before calling the function, the subprocess starts a debugpy[9] listener on `localhost:5678` and waits for a debugger to attach; the test log shows:

```
py_func waiting for the debugger on localhost:5678. Attach from your IDE,␣
↪or Stop to cancel
```

Requirements: install debugpy with the host Python, the `python_bin` interpreter, the same one running your functions:

```
<python_bin> -m pip install debugpy
```

Then attach from the IDE. VSCode configuration (`launch.json`):

```
{
    "name": "Attach to py_func",
    "type": "debugpy",
    "request": "attach",
    "connect": {"host": "localhost", "port": 5678},
    "justMyCode": true
}
```

Set breakpoints in your `.py` file and press F5: once attached, the function runs and stops on your breakpoints. Notes:

- the run resumes only after the debugger is attached. Stop cancels the wait (the item fails); an item left waiting fails after one hour;
- if debugpy is missing or the port cannot be opened, the function is not run: the item fails with an explanatory message plus a `WARN` line, and the test run continues;
- the listening port can be changed with the `py_func_debug_port` global variable (configuration file or `-d py_func_debug_port=<n>`). If the port is busy, the next free one (up to 19 above) is used and announced in the test log;

[9] https://github.com/microsoft/debugpy

- with `context_id`, the shared subprocess opens the listener once: the first `debug: true` item fixes the port (a later change is ignored with a warning) and the debugger stays attached for the following items. If the IDE disconnects, the next `debug: true` item waits for a new attach;
- the listener only accepts local connections. To debug a testium running on a remote machine, open an SSH tunnel: `ssh -L 5678:localhost:5678 <remote>`;
- with the `redirectOutput` debugger option, the function's prints may appear both in the IDE debug console and in the testium log.

Python Interpreter environment setup

Some global variables have an impact on the `py_func` test item behavior:

- `python_bin`: This optional global variable can be used to define the python executable path. If not defined, the python interpreter is looked up in the default system locations.
- `python_env`: This global variable can be used to define environment variables for the python script execution environment. Only *PATH* and *PYTHONPATH* are supported.

Listing 4.53: example of configuration file: param.yaml

```
[...]
python_env:
  PATH: "/my/path/"
  PYTHONPATH: "/my/python/modules/"
[...]
```

4.3.21 pytest test item

The `pytest` test item runs a pytest[10] test file and turns every collected test into a child item, each with its own PASS / FAIL / SKIP result, duration and failure message. It is the pytest counterpart of the `unittest` test item.

The tum file prototype is as follows:

Listing 4.54: `pytest` test item usage example

```
- pytest:
    name: pytest test item
    test_file: test_device.py
    test_method:
        - test_boot
        - test_version
```

Attributes

Beside common test items attributes, the pytest test item has specific attributes:

- `test_file`: the name (and optionally the path) of the pytest file to run.
- `test_method`: optional list of test function names. When present, only the matching tests are included in the test tree (the name is matched against the function part of each pytest *node id*, the parametrisation suffix being ignored). Otherwise every collected test in the file is run.

[10] https://docs.pytest.org

Host execution

Unlike `unittest` (which runs in *testium*'s own interpreter), the `pytest` item runs pytest in a subprocess on the host interpreter, the same one used by `py_func` / `lua_func` (overridable with the `python_bin` global variable). As a consequence:

- `pytest` (and the test's own dependencies) must be installed on that host interpreter, e.g. `pip install pytest`. It is not bundled with *testium*.
- the tests run isolated from *testium*'s in-process API; they are meant to be self-contained (they exercise the device/software under test, not the *testium* tree).

The item is invoked with a controlled pytest configuration (`--capture=no`, user addopts neutralised, cache disabled); a small built-in pytest plugin reports the collected tests and their results back to *testium*.

4.3.22 report test item

This test item exports report files **during** the test run, from the data collected so far. It snapshots partial results in a long campaign.

It requires a `report` element at the root of the main test file (see the *Reports chapter*).

Listing 4.55: `report` test item usage example

```
- report:
    name: Intermediate report
    export:
        - junit:
            path: $(home)/reports
            file_name: intermediate.xml
            pattern:
                - Unittest%
        - text:
            path: $(home)/reports
            file_name: report-key-1.txt
            key:
                - report-key-1
```

Attributes

Besides the *common attributes*, the `report` test item has one specific attribute:

- `export`: required. One export entry or a list of them, with exactly the same syntax and attributes (`path`, `file_name`, `pattern`, `key`, `cmd`) as the root-level `export`; see *export attributes*. All formats are available: *built-ins*, the *command export* and installed *plugins*.

Notes:

- a `sqlite` entry is ignored here: the database storage is configured once, in the root `report` element;
- exports run with `no_header` set: the produced files carry the test rows without the run header (the run is not finished, so the global result and duration do not exist yet).

4.3.23 run test item

This test item executes a new instance of testium with the specified `.tum` file.

- In **batch mode** (`-b`): the sub-instance is started with `-b`.
- In **GUI mode**: the sub-instance opens its own window with `-r` (run and close).

- batch: true forces the sub-instance to run headless (-b) even in the GUI.

A sub-instance started with -b is **captured**: its output is streamed into this test's log and report, and kept as the item's result value, so it can be stored with store_result and post-processed (expected_result, process_result, or a py_func reading the global variable).

The item result is **PASS** if the sub-instance launched and ran to completion, regardless of whether the sub-tests passed or failed. It is **FAIL** if the file could not be found, the sub-instance could not be launched, or the time window was not reached (see start_time / end_time).

Listing 4.56: run test item usage example

```
- run:
    name: Execute TUM
    tum: example_cycle.tum
    log_file: $(home)/reports/test.log
    report_file: $(home)/reports/test.rep
```

Attributes

run test item has the following specific attributes:

- tum: mandatory, the path of the file to execute. Can be relative to the current execution folder.
- param_file (optional): the path of the parameter file to use; otherwise the default parameter file is used.
- batch (optional): true to run the sub-instance headless (-b) and capture its output even in GUI mode (see above).
- log_file (optional): the path of the log file. In GUI mode, if not provided, a file is created with a timestamp next to the .tum file. Not used in batch mode (the output is captured instead).
- report_file (optional): the path of the report file to create.
- start_time (optional): earliest time to execute the sub-instance, in HH:MM format.
- end_time (optional): latest time for execution within a time frame, in HH:MM format.
- wait_for_exec (optional): true to wait until the time window defined by start_time and end_time is reached before running. Requires both start_time and end_time.

4.3.24 sleep test item

sleep test item has the following description format

Listing 4.57: sleep test item example usage

```
- sleep:
    name: sleep test item
    timeout: 10
    dialog: True
```

Attributes

- timeout: sleep duration, see *Duration format*.
- dialog: If set to True, a window showing the remaining time to wait is displayed (optional parameter set to False by default)

Duration format

A duration is a number of seconds, or a string with units.

Listing 4.58: Duration examples

```
timeout: 10
timeout: 1.5
timeout: 1m 30s
timeout: 2d 5h 31m 3s
timeout: 1.5h
```

- Units are `d` (days), `h` (hours), `m` (minutes) and `s` (seconds), in this order.
- Each unit is optional. Spaces between units are optional.
- Decimal values are accepted. Negative values are refused.
- Any other text fails the item: `1h30` and `5m 3h` are errors.

The same format applies to the `period` of the *plot periodic action*.

4.3.25 unittest test item

The unittest test item runs a test script written with the `unittest` module from the python standard library.

The tum file prototype is as follows:

Listing 4.59: `unittest` test item usage example

```
- unittest:
    name: unitTest test item
    test_file: unitTestScript.py
    test_method:
        - test_1
        - test_2
```

Attributes

Besides the common test item attributes, the unittest test item has specific attributes; `test_file` is mandatory.

- `test_file`: it is the name (and optionally the path) of the unittest file to be processed.
- `test_method`: it is an optional unittest test sub-item. If one or more elements are present, the unittest python script file is parsed and only the corresponding methods are included in the test tree. Otherwise, all the test methods are included in the test tree.

Access to global variables entries

`unittest` file tests instances have access to the testium global variables by using the *helper's library*.

Report value from unittest

Values can be added to the test report from a `unittest` test at runtime.

Listing 4.60: example of `unittest` test item python function

```
from  unittest import (TestCase)

class DummyTests(TestCase):
    def test_01_report(self):
        self.reported_values['key reported']= 'value_reported'
```

Console use example with unittest item

Here is an example how to use the console module from python `unittest`.

Listing 4.61: example of a *testium* console usage from a `unittest` python function

```
from  unittest import (TestCase)
import console

class DummyTests(TestCase):
    @classmethod
    def setUpClass(cls):
        cls.consA0= console.TelnetConsole('cons name','192.168.98.123',
↪7001)
        cls.consA0.open()
        cls.promptA0 = 'test-computer>'

    def test_01_console(self):
        self.consA0.write('config')
        self.assertEqual(self.consA0.read_until(self.promptA0, 10), 0)
        self.consA0.write('lsusb && echo "Done."\n')
        status, read_data = self.consA0.read_until('Done.',
                                                   10, return_data=True)
        self.assertEqual(status, 0)
        if read_data.find('ID 04f2:b684 Chicony Electronics Co.')!=-1:
            index=0

    @classmethod
    def tearDownClass(cls):
        cls.consA0.close()
```

4.4 Includes

It is possible to include one TUM file in another by using the `!include` tag before the file name.

This feature is a testium specific implementation and is not part of the YAML language, although it is based on the tagging feature of the language and the customization possibility offered by the python pyYaml package.

Here is a basic example of file inclusion:

Listing 4.62: included_file.tum

```
- test_item:
    name: test_2
```

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```
- test_item:
    name: test_3
```

Listing 4.63: main.tum

```
#include with the sub-sequence reference mechanism
sequence: &included_sequence
    !include included_file.tum

main:
    name: Test example
    steps:
        - test_item1:
            name: test_1
        - *included_sequence

        #include can also be inserted directly within the steps list
        - !include included_file.tum
```

An include can also carry arguments for the included file:

```
- !include {file: included_file.tum, sensor: sensor_A}
```

The arguments feed the jinja template of the included file; their resolution time is described in *templates*.

4.5 Templates

testium embeds the jinja2[11] template engine. It allows extensive customization of test files and makes test scripts reusable.

Templates are rendered at load time.

4.5.1 In the main test file

The *testium* main test files are always passed through the jinja template engine.

The variables available to jinja are the *configuration file* entries plus the *built-in values*. Jinja receives their resolved values, as they are at load time.

4.5.2 In `!include` directive

In addition to basic inclusion, `!include` accepts arguments. These arguments replace the corresponding `{{ keyword }}` placeholders in the included file.

Listing 4.64: including a template

```
main:
    name: Test example
    steps:
        - test_item1:
            name: test_1

        #include can also be inserted directly within the steps list
```

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[11] https://jinja.palletsprojects.com

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```
        - !include
            file: included_template_file.tum
            inclusion_parameter_1: param1
            inclusion_parameter_2: param2
```

Listing 4.65: included template

```
- test_item:
    name: {{ inclusion_parameter_1 }}
- {{ inclusion_parameter_2 }}:
    name: test_3
```

4.5.3 Include arguments and resolution time

Listing 4.66: calling file

```
- !include {file: bench.tum, cfg: $(bench_config)}
```

Listing 4.67: bench.tum

```
{% set _cfg = expand(cfg) %}
{% if _cfg.mode == "fast" %}
- sleep: {name: warmup, timeout: 0.1}
{% endif %}

- py_func:
    name: run bench
    file: bench.py
    func_name: run
    param:
        - {{ cfg }}
```

- An argument is passed to the template as declared: `{{ cfg }}` writes the text `$(bench_config)` into the file.
- That text is resolved when the item runs, with the value in force at that moment. A later redefinition of `bench_config` applies.
- `expand(value)` resolves `$( )` and `<| |>` at load time. Use it to access the value inside the template itself: attribute access, `{% if %}`, `{% for %}`. `expand()` exists only in templates.
- Mixing the two syntaxes in one expression is not allowed: `{{ $(inclusion)_parameter }}` fails to load. Write the `$( )` part as text and let it resolve at run time.

4.6 Reports

During a test run, testium records one row per executed test item (name, type, result, message, duration, reported values) into a SQLite database, together with a header describing the run: test file, date, global result. A report export turns that database into an output file: JUnit XML, HTML, JSON, plain text, the raw database itself, or any custom format.

This chapter describes:

- how to enable the report and declare exports (*declaring exports*),

- the attributes accepted by every export entry (*export attributes*),
- the built-in formats and the special role of `sqlite` (*built-in formats*),
- post-processing with an external program (*command export*),
- adding custom formats as pip plugins (*exporter plugins*),
- the report database schema (*database schema*).

4.6.1 Declaring exports

The `report` element at the root of the main `.tum` file enables the recording and lists the exports. The `export` key takes one entry or a list of entries; each entry uses the format name as its key:

Listing 4.68: report declaration with multiple exports

```
report:
    enabled: True
    log_stored: True
    export:
        - sqlite:
            path: $(home)/reports
            file_name: $(test_name).db
        - junit:
            path: $(home)/reports
            file_name: $(test_name).xml
        - html:
            path: $(home)/reports
            file_name: $(test_name).html
```

Table 4.4: report attributes

Attribute	default value	Description
`enabled`	`True`	Enables report recording.
`log_stored`	`False`	When `True`, the stdout of each test item is captured during the run, so that exports (`html`, `json`) can include the log of each item.
`export`	/	One export entry or a list of them. Each entry's key is the format name (see *Built-in formats*, *command export: external tool*, *Custom export formats (plugins)*).

When the exports run:

- At the end of the run: each entry of the `export` list produces one output file, in declaration order. This is the normal case.
- During the run: a *report test item* in the test sequence runs its own `export` list on the data collected so far. It snapshots partial results in a long campaign.
- `sqlite` is the exception: it is not converted at the end but written live during the run (see *Built-in formats*).

In *batch mode*, the `-p`, `-t` and `-n` command-line options (see *-p*) replace the `export` list of the test file with a single export built from the given path, type and patterns.

4.6.2 Export attributes

Every export entry accepts the following sub-attributes:

Table 4.5: export attributes

Attribute	default value	Description
`path`	`$(report_path)`	Output directory.
`file_name`	/	Output file name.
`pattern`	/	One or more SQL LIKE patterns applied on the test item `name`.
`key`	/	One or more SQL LIKE patterns applied on the test item key attribute.
`cmd`	/	`command` export only: the external command line to run (see *command export: external tool*).

`path` and `file_name` are joined to build the output file; you can also put the full path in just one of the two attributes. All attributes accept `$(...)` *global variable expansions*, resolved at export time. For example `file_name: $(test_name).xml` produces one report file per test file. If the output file already exists, it is renamed to `<name>-<N>.saved` first, never overwritten.

`pattern` and `key` select which test items appear in the output. Without them, every executed item is exported. Both use the SQL LIKE syntax: `%` matches any text, `_` matches one character. They accept a single string or a list; an item is kept when it matches any of the given patterns.

- `pattern` matches on the item `name`;
- `key` matches on the item key attribute, one of the *common test item attributes*. The usual scheme is to tag related items with the same key in the `.tum` and filter the report on it:

Listing 4.69: selecting items with key

```
steps:
    - console:
        name: power on
        key: setup
        ...

report:
    export:
        - junit:
            file_name: setup_only.xml
            key: setup
```

4.6.3 Built-in formats

- `sqlite`: the report database itself (see below).
- `text`: indented text dump of the test tree.
- `json`: full report as JSON: `{"header": {...}, "tests": [...]}`.
- `junit`: JUnit XML, for CI systems (requires the `junit_xml` Python package).
- `html`: single HTML page with header, results table and per-item logs (requires `lxml`).

`sqlite` has a special role: it is not a conversion but the storage layer itself. With a `sqlite` entry, the internal database is written live to the given file during the run; the file remains

afterwards for any external analysis (schema in *Report database schema*). Without it, the database only lives in memory and disappears once the other exports have run. Declaring several `sqlite` entries is pointless: only the last `path` is used.

The `html` and `json` exports can embed the captured stdout of each test item; this requires `log_stored: True` in the `report` block (see *Declaring exports*).

If a format is unknown, or an optional dependency is missing, the export is skipped with an `[report] Export skipped: ...` line on stdout and the test run continues. The line lists the available formats: built-ins, `command`, plus every installed plugin.

4.6.4 `command` export: external tool

The `command` export post-processes the report with any external program. The report database is copied to a temporary SQLite file and the command runs on the host system (even from the Flatpak / AppImage sandboxes):

Listing 4.70: post-processing the report with an external tool

```
report:
    export:
        - command:
            cmd: mytool --input $(db) --output $(out)
            path: $(home)/reports
            file_name: $(test_name).pdf
```

Two placeholders, following the usual `$(...)` syntax, are reserved in `cmd` (they are resolved by the export itself, not from the global variables):

- `$(db)`: path of the temporary SQLite copy of the report (schema in *Report database schema*);
- `$(out)`: the output path built from `path` / `file_name`.

`cmd` is split shell-style (quotes group words) but not run through a shell: no pipes, redirections or globbing. Wrap them in a script if needed. Other `$(...)` *expansions* work as usual, so `$(python_bin) myscript.py $(db) $(out)` runs a script with the same interpreter as *py_func*. The command runs with the test directory as working directory, so relative paths behave as elsewhere in the `.tum` file. The temporary `$(db)` copy is deleted after the command returns; the command must not rely on it afterwards.

The command output (stdout) is forwarded to the test log. A command that cannot be started or exits with a non-zero code produces an `[report] Export skipped: ...` line (with its stderr); the test run continues.

4.6.5 Custom export formats (plugins)

A third-party Python package can register additional export formats via the `testium.exporters` setuptools entry point group. A step-by-step tutorial (package layout, install, use) is provided in `doc/exporter_tutorial.md` of the source repository; this section is the reference.

The plugin must be installed with the host Python interpreter, the one resolved as `python_bin` and used by *py_func*. A plain `pip install` with that interpreter is enough; no testium configuration change is needed. The format is then usable in any `.tum` by its declared name, in every install channel: source, wheel, PyInstaller, Flatpak and AppImage.

Execution model: any format name that is not a built-in is looked up on the host. testium copies the report database to a temporary SQLite file, spawns the host Python, scans the installed `testium.exporters` entry points there, loads the matching class and instantiates it with the database copy. Consequences:

- the plugin (and any dependency it imports) must be installed with `python_bin`'s `pip`, not inside the testium bundle (read-only in the frozen channels);
- the plugin cannot use testium internals: it only sees the SQLite database (*Report database schema*) and its constructor arguments;
- a crash in the plugin (or a missing plugin dependency) skips that export with an `[report] Export skipped: ...` line and the run continues.

The plugin declares its format name in its `pyproject.toml`:

Listing 4.71: registering an exporter via entry-points

```
[project.entry-points."testium.exporters"]
my_format = "my_pkg:MyExporter"
```

The `testium_report` helper

The recommended way to write an exporter is the `testium_report` helper module shipped with testium (the built-in formats are implemented with it). The plugin imports it with no installation step: the process loading the plugin class resolves it from the running testium, so its version always matches. Subclass `Exporter` and implement `export()`:

Listing 4.72: exporter built on the helper

```
from testium_report import Exporter

class MyExporter(Exporter):
    def export(self):
        # self.rows      : filtered items; .name .type .key .result
        #                  (.passed/.failed/.skipped), .message,
        #                  .duration_s, .level, .log, .data (decoded)
        # self.report    : .header dict, .rows(pats, keys), .tree()
        # self.out_path  : output path (previous file renamed .saved)
        # self.name / self.no_header : report name, inline-item call
        with open(self.out_path, "w") as f:
            for row in self.rows:
                f.write(f"{row.name}: {row.result}\n")
```

`Report` also accepts a database file path instead of a connection, which makes exporters easy to unit-test against a saved `sqlite` report. For the development environment (IDE, tests), `pip install testium-hil` provides the same `testium_report` module as a regular top-level import; at execution under testium the shipped copy takes precedence.

Writing an exporter without the helper

The helper is optional. To run an export, testium only instantiates the plugin class with the arguments below. A class with this constructor can read the SQLite tables (*Report database schema*) directly, without `testium_report`:

Listing 4.73: minimal exporter class

```
class MyExporter:
    def __init__(self, name, con, path, pats, keys, no_header=False):
        # name      : str; report name
        # con       : sqlite3.Connection (read); tables: header, tests
        # path      : str; output file path (variables already expanded)
        # pats      : list[str]; LIKE filters on test_name (may be empty)
        # keys      : list[str]; LIKE filters on report_key (may be empty)
```

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```
        # no_header : bool; do not write the run header (set when the
        #             export comes from an inline `report` test item)
        ...  # do the work in __init__ and write to `path`
```

4.6.6 Report database schema

The `sqlite` export produces this database; `command` exports and plugins receive a temporary copy. It holds two tables.

`header`: one (`key TEXT, value TEXT`) row per run property: `report_version`, `test_file`, `test_name`, `test_result`, `test_revision`, `testium_version`, `testrun_date`, `testrun_time`, `test_duration`.

`tests`: one row per executed test item, 12 columns:

Table 4.6: `tests` table columns

Column	Content
`timestamp_start`	Start timestamp; the table is ordered on it.
`test_id`	Unique item id.
`parent_id`	`test_id` of the enclosing item (tree structure).
`level`	Nesting depth.
`test_name`	Item `name` attribute.
`test_type`	Item type (`Console`, `Sleep`, ...).
`report_key`	Item key attribute (see *Items common attributes*).
`result`	`PASS`, `FAIL` or `SKIP`.
`message`	Result message.
`duration`	Item duration (0.1 ms units).
`log`	Captured stdout of the item, when `log_stored: True`.
`data`	JSON of the values reported by the item, e.g. `reportValue(...)` in a *py_func*.

4.7 Test outputs

A list of test result outputs is automatically updated by *testium*.

This is an entry of the global variables dataset whose key is `test_outputs`.

This global_dict member contains the log file path and, if configured, the report path as a list.

The user can add other logged files by updating this global variables entry.

4.8 Post execution

A post execution script can be run for example to copy the output files.

For that, a `post_execution` element can be defined in the .tum file.

If the test run succeeds, the `post_exec` function of the `file_name` module is run; otherwise `post_exec_fail` is run.

If the post_execution element is not defined, the post_execution.py file in the test directory is used by default if it exists.

Listing 4.74: custom post execution python file

```
post_execution:
    file_name: test_report_text.py
```

4.9 Sub-sequence references

It is possible to alias any part of the TUM description file (typically a sequence of steps to be executed) to be inserted within another sequence.

This feature uses the anchor/alias mechanism of the YAML language[12].

Here is an implementation example of a reference to a sub-sequence in a TUM file:

Listing 4.75: sub-sequences call

```
sequence: &temperature_step_sequence
    - test_item:
        name: test_2
    - test_item:
        name: test_3

main:
    name: Test example
    steps:
        - test_item1:
            name: test_1
        - *temperature_step_sequence
```

Note

The entry before the alias (sequence: in the example above) is required by YAML syntax. However, its value is not used by *testium* and can be anything.

4.10 Test documentation

It is possible to display explanatory text in the GUI.

The doc attribute of test items is used for that purpose and is displayed as a tooltip on the test row.

Listing 4.76: tests documentation

```
main:
    name: Test example
    steps:

        - unittest:
            name: unittest item
            doc: |
                The purpose of this unittest test item is to demonstrate
                its various features.
```

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[12] https://yaml.org/

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```
            test_file: dummy/dummy.py
            test_method: test_01_pass
```

See illustration in Figure 4.3.

4.10.1 Unittest

For `unittest` type test items, the python docstring of the test method is used as documentation.

4.11 Running several testium instances

Several testium instances can run at the same time on the same machine.

- Settings are merged per key when an instance closes: each instance only writes back what it changed.
- The GUI keeps the tree state (folds, checkboxes, breakpoints) and the log file choice per test file, for the last 20 files opened.
- Output paths written in the `.tum` (reports, plot files, `json_rpc` UDP reception ports) are not made unique: tests run in parallel must use distinct values.
- If the `py_func` debug port is busy, the next free one is used and announced in the test log.

4.12 Window layout

The Log, Test doc, Step info and Variables panels are dockable: drag a panel by its title bar to any side, below the tree, below the log, or drop it on another panel to tab them. Panels close from their title bar; the View menu lists them and can reset the layout. The arrangement is saved on exit, also during a run.

The tree header holds a follow button (the tree scrolls with the running step; scrolling by hand disengages) and a magnifier opening the step search. The Result column shows the last message on one line: click it for the dated failure history, hover it for the full text.

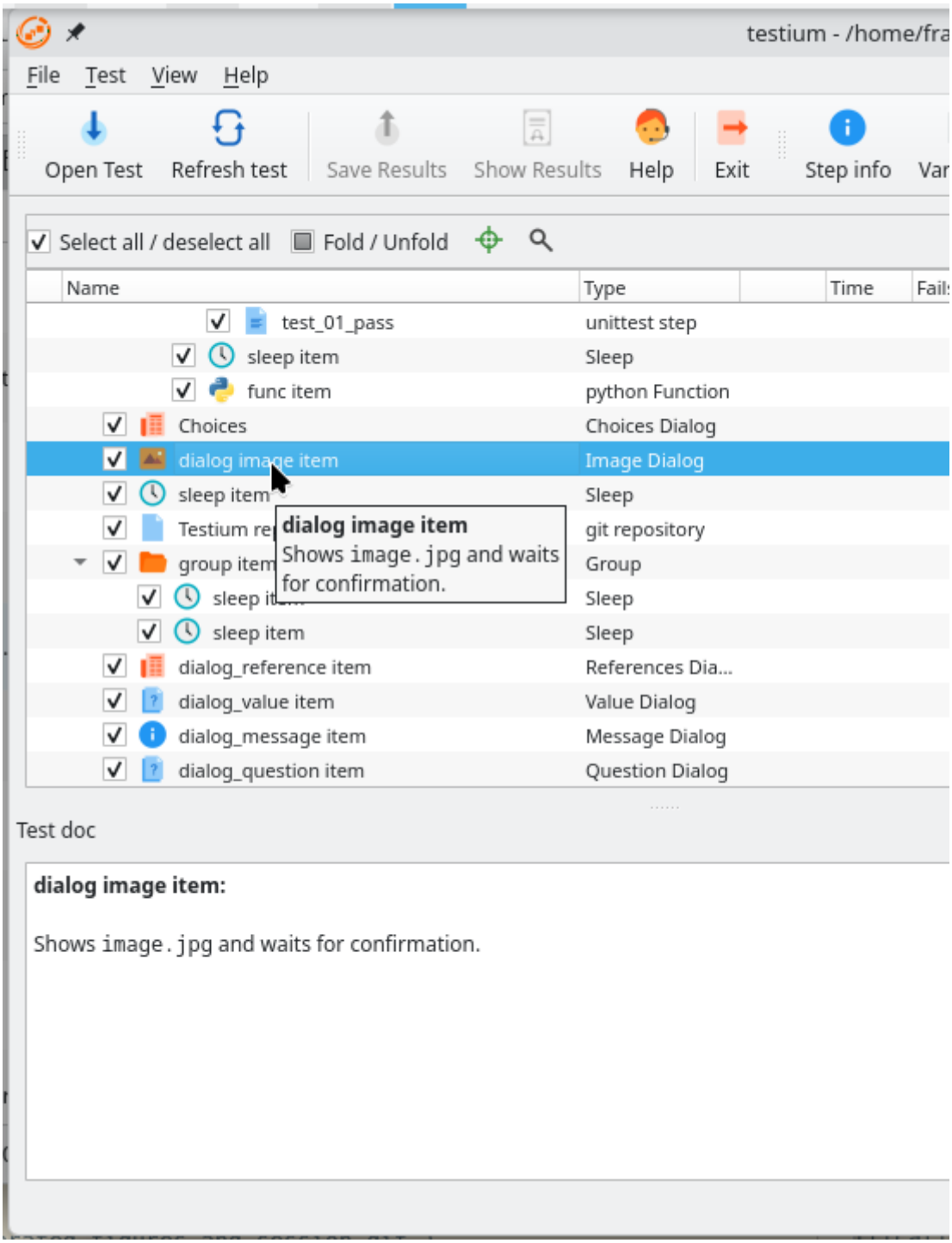


Fig. 4.3: Illustration of the doc attribute effect in the GUI.

CHAPTER

FIVE

PYTHON HELPER LIBRARY

A Python library of helper functions for Python modules called from testium `py_func` items.

User scripts run inside the `py_func` subprocess and interact with testium through a JSON-RPC bridge, the `py_func.tm` module. They must not import `api.testium` or `interpreter.*` directly: those are main-process modules and may not even be reachable in a packaged build (PyInstaller, .deb).

To include the support of this library in a python script, the following line must be included in the script header:

Listing 5.1: testium helper library import

```
import py_func.tm as tm
```

5.1 Global variables helper functions

To manage values in the global variables dataset:

delgd(*name*)

Remove an entry from the testium global dictionary.

Parameters
name (*str*) – Name of the entry to remove.

gd(*name*, *default=None*)

Return a value from the testium global dictionary.

A value declared with `$(  )` or `<|  |>` is returned resolved, with the variables in force at call time.

The value is accessible from any test item and from any `py_func` subprocess, regardless of the `context_id` used.

Parameters

- **name** (*str*) – Name of the entry to retrieve.
- **default** – Value returned when the key is absent. Defaults to `None`.

Returns
The stored value, or *default* if not found.

setgd(*name*, *value*)

Store a value in the testium global dictionary.

The stored value is accessible from any subsequent test item and from any `py_func` subprocess via `gd()`.

When `context_id` is used on the `py_func` item, any Python object (including those that cannot be transmitted to other processes) can be stored and shared between calls running in the same subprocess.

Parameters

- **name** (*str*) – Name of the entry to set.
- **value** – Value to store.

5.2 Plot helper functions

Add values to a running plot or read the last value from it:

add_plot_values(**params*)

last_plot_value(**params*)

Console and plot **lifecycle** management (`add_console`, `remove_console`, `console`, `add_plot`, `remove_plot`, `plot`) is performed by the `console` and `plot` test items themselves, not from user `py_func` scripts. Use those test items to open/close consoles and plots.

5.3 Other helper functions

OS(**params*)

get_main_dir(**params*)

init_timestamp(**params*)

text_mode(**params*)

timestamp(**params*)

timestamp_as_sec(**params*)

5.4 Debug mode

The `test_debug` global variable controls debug-only output. Read or write it via `tm.gd("test_debug")` / `tm.setgd("test_debug", True)`. See *Debugging your tests*.

CHAPTER

SIX

DEBUGGING YOUR TESTS

testium provides four complementary tools to debug a test campaign: breakpoints and step-by-step execution to control the flow, the Variables panel to inspect the data, the Debug output switch for verbose diagnostics, and an IDE debugger attach for the Python code of `py_func` items.

6.1 Breakpoints

Double-click in the first column of the test tree (left of the checkboxes), or select the item and press **Space**, to set or remove a breakpoint, shown as a red dot. The run pauses before executing a marked item. Breakpoints follow the item across Refresh and testium restarts, and are stored per test file.

Right-click an item and pick **Breakpoint condition** to pause only when an expression is true. Plain Python with `$(var)` substitution; the `<| ... |>` markers are not needed. A conditional breakpoint is shown as a red ring; the expression appears in its tooltip. An expression that fails to evaluate pauses anyway, with a WARN.

6.2 Step-by-step execution

The Run toolbar (also the Test menu) drives the execution. The step buttons and their shortcuts appear when the **Debug** button is enabled:

- **F5**: start, pause or resume. Pause lets the current step finish and stops the run before the next one; **Shift+F5**: stop;
- **F10**: step over, run to the next item at the same level;
- **F11**: step into, run to the next item at any depth. From idle, it starts the run paused on the first item;
- **Shift+F11**: step out, run to the end of the current level, pause one level up;
- **Shift+F10**: re-run last step, re-execute the step that just ran and pause on the next one. On an item holding actions (console, plot, json_rpc), the whole item is re-executed.

While paused, the green line marks the current item.

6.3 Jump to a step

While paused, right-click any item and pick **Jump to this item**: execution moves there, forward or backward, and pauses on it. Nothing in between is executed; on a backward jump the variables keep their current values. Jumping into or out of a `parallel` block is refused.

6.4 Editing the run

The checkboxes stay operative while a test runs: unchecking an item skips it when its turn comes. The item currently executing is not affected.

6.5 Inspecting variables

F1 shows the Variables panel: the global dictionary, filterable, editable while the test is paused or between runs. Editing a variable also changes the values derived from it (see *when values are resolved*). The panel is a dock like Log and Test doc: it can be closed, moved to any side, placed below the tree or tabbed; the View menu and the toolbar button bring it back.

The expression field below the table evaluates a Python expression against the current variables, also while paused on a breakpoint. `$(var)` is substituted; the `<| ... |>` markers are not needed; a lone `$(var)` shows the value without evaluating it. It shows the result, or the error when the expression is invalid. The text is taken as-is (no YAML quoting rules).

6.6 Debug output

The **Debug** button of the Run toolbar (**Ctrl+Shift+D**) enables the verbose diagnostics: the `DEBUG` log lines, including the remedy hints attached to some warnings. It drives the `test_debug` global variable, takes effect immediately, even during a run, and keeps its value between runs and between testium sessions.

A `test_debug` value set in a configuration file or with `-d test_debug` overrides the button state at load; the button then shows the effective value. In batch mode, use `-d test_debug`.

6.7 Debugging `py_func` code

The Python functions of `py_func` items can be debugged from your IDE (breakpoints in your `.py` file, stepping, variable inspection). Two ways to arm it:

- right-click the `py_func` item in the tree → **Wait for IDE debugger**. A blue dot marks the item; the request lasts for the session only (a Refresh clears it);
- or set `debug: true` on the item in the `.tum` file.

At the next run, the item waits for a debugger to attach on `localhost:5678` (global `py_func_debug_port`; if busy, the next free port is used and announced in the log) before calling the function. Details and IDE configuration: the *py_func chapter* and the step-by-step tutorial `doc/debug_tutorial.md` of the source repository.